Linear Algebra I, Sheet 2, MT2026

More about matrices. Elementary row operations; echelon form of a matrix.

Main course

1. Let J_n be the $n \times n$ matrix with all entries equal to 1. For what values of α, $\beta \in \mathbb{R}$ is the matrix $\alpha I_n + \beta J_n$ is invertible?
[Hint: note that $J_n^2 = nJ_n$; seek an inverse of $\alpha I_n + \beta J_n$ of the form $\lambda I_n + \mu J_n$ where λ, $\mu \in \mathbb{R}$.]

Find the inverse of $\begin{pmatrix} 3 & 2 & 2 & 2 \\ 2 & 3 & 2 & 2 \\ 2 & 2 & 3 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}$.

2. Use EROs to reduce each of the following matrices to RRE form:

$$\text{(a)} \begin{pmatrix} 2 & 4 & -3 & 0 \\ 1 & -4 & 3 & 0 \\ 3 & -5 & 2 & 1 \end{pmatrix}; \quad \text{(b)} \begin{pmatrix} 1 & 2 & 3 & 0 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 5 & 2 \end{pmatrix}; \quad \text{(c)} \begin{pmatrix} 1 & 2 & 3 & 0 \\ 2 & 3 & 4 & 2 \\ 3 & 4 & 5 & 2 \end{pmatrix}.$$

3. Let A and B be $m \times n$ matrices, and let C be an $n \times p$ matrix.
(i) Show that $(A+B)^T = A^T + B^T$ and that $(\lambda A)^T = \lambda A^T$ for scalars λ.
(ii) Show that $(AC)^T = C^T A^T$.
(iii) Suppose that $m = n$ and A^{-1} exists. Show that A^T is invertible and that $(A^T)^{-1} = (A^{-1})^T$.

4. The *trace* of a square matrix is the sum of its diagonal elements. Let A be an $m \times n$ matrix and B be $n \times m$. Show that

$$\text{trace}(AB) = \text{trace}(BA).$$

5. Use EROs to find the inverses of each of the following matrices

$$\begin{pmatrix} 2 & 3 \\ 3 & 2 \end{pmatrix}; \qquad \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}; \qquad \begin{pmatrix} 1 & -1 & 0 & 0 \\ 1 & 0 & -1 & 0 \\ 1 & 0 & 0 & -1 \\ 0 & 1 & 1 & 1 \end{pmatrix}.$$

6. (i) Show that if the $m \times n$ matrices A, B can be reduced to the same matrix E in echelon form, then there is a sequence of EROs that changes A into B.

(ii) Show that an $n \times n$ real matrix may be reduced to RRE form by a sequence of at most n^2 EROs.

Starter

S1. Use EROs to determine whether the following matrices are invertible. For each invertible matrix, find the inverse.

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}; \qquad \begin{pmatrix} 1 & 2 & 3 \\ -1 & 0 & 1 \\ 2 & 4 & 6 \end{pmatrix}; \qquad \begin{pmatrix} -2 & 0 & 4 & 3 \\ 1 & 7 & 5 & -6 \\ -3 & 7 & 13 & 0 \\ 0 & 1 & 2 & 3 \end{pmatrix}.$$

S2. Show that the inverse of an ERO is an ERO. (Hint: consider each of the three types of ERO separately.)

S3. Prove from the vector space axioms that if V is a vector space, v, $z \in V$ and $v + z = v$, then $z = 0_V$.

Pudding

P1. Use EROs to explore for which real numbers a, b, c, d the 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is invertible. You will need to proceed on a case by case basis.

P2.

(i) What happens if we multiply an upper triangular matrix by an upper triangular matrix?
(ii) What if we multiply a lower triangular matrix by a lower triangular matrix?
(iii) What if we multiply a lower triangular matrix by an upper triangular matrix?

P3. Let $V := \mathbb{R} \times \mathbb{Z}$, the set of all pairs (x, k) where x is a real number and k is an integer. Define addition coordinatewise so that $(x, k) + (y, m) = (x + y, k + m)$, and define scalar multiplication by real numbers λ by the rule $\lambda(x, k) = (\lambda x, 0)$. Which of the vector space axioms are satisfied, and which are not?